DEEP LEARNING

Unit – 3

1. Execute an example function for Convolution operation and explain in detail. (13marks)

Convolution Operation in Convolutional Neural Networks (CNN)

Definition

The **convolution operation** is the fundamental operation in CNNs used to extract important features from input data such as images.
It involves sliding a small matrix called a **kernel or filter** over the input matrix and computing the dot product between the filter and local regions of the input.

$$S(i,j) = (I * K)(i,j) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} I(i+m, j+n) \cdot K(m,n)$$

- $I(i,j)$: The pixel value at position $I(i,j)$ in the input image.
- $K(m,n)$: The weight of the kernel at position (m,n).
- $S(i,j)$: The output feature map at position (i,j).

Example of Convolution Operation

Consider a **5 × 5 input image matrix** and a **3 × 3 filter**.

Input Image

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

Filter / Kernel

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Step-by-Step Convolution Process

Step 1: Select First Region

Take the **top-left 3×3 region** of the image.

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 2: Element-wise Multiplication

Multiply corresponding elements of the image region and kernel.

$$(1 \times 1) + (1 \times 0) + (1 \times 1) + (0 \times 0) + (1 \times 1) + (1 \times 0) + (0 \times 1) + (0 \times 0) + (1 \times 1)$$

Step 3: Summation

$$1 + 0 + 1 + 0 + 1 + 0 + 0 + 0 + 1 = 4$$

So the **first value of the feature map = 4**

Step 4: Slide the Filter

The filter moves across the image **one step at a time (stride = 1)**.

The same calculation is repeated for all regions until the entire image is covered.

Output Feature Map

After applying convolution to the entire input image, the output becomes:

$$\begin{bmatrix} 4 & 3 & 4 \\ 2 & 4 & 3 \\ 2 & 3 & 4 \end{bmatrix}$$

This resulting matrix is called the **feature map** or **activation map**.

Important Parameters in Convolution

1. Kernel (Filter)

A small matrix used to detect specific features like edges or textures.

2. Stride

Stride is the number of steps the filter moves each time across the image.

3. Padding

Padding adds extra pixels around the input image to maintain the spatial size of the output.

4. Feature Map

The output produced after applying convolution is called the feature map.

Advantages of Convolution Operation

1. **Automatic Feature Extraction**
   CNN automatically learns important features from images.

2. **Parameter Sharing**
   The same filter is applied across the entire image, reducing parameters.
3. **Sparse Connectivity**
   Each neuron connects only to a small region of the input.
4. **Translation Invariance**
   CNN can detect objects even if they appear in different positions.

Applications

Convolution operations are widely used in:

- Image classification
- Object detection
- Face recognition
- Medical image analysis
- Autonomous driving systems

2. Examine Pooling with suitable example. (13marks)

Pooling in Convolutional Neural Networks (CNN)

Definition

Pooling is a **down-sampling operation** used in Convolutional Neural Networks to reduce the spatial size of feature maps.
It helps decrease computational complexity, reduce parameters, and prevent overfitting while preserving important features.

Purpose of Pooling

The main objectives of pooling are:

1. **Reduce Feature Map Size** – decreases width and height of feature maps.
2. **Reduce Computation** – fewer parameters lead to faster training.
3. **Feature Invariance** – helps detect features even if their position slightly changes.
4. **Noise Reduction** – keeps dominant features while removing unnecessary details.

Types of Pooling

1. Max Pooling

Max pooling selects the **maximum value** from a region of the feature map.

2. Average Pooling

Average pooling computes the **average value** of the elements in the region.

3. Global Pooling

Reduces the entire feature map into a **single value per feature map**.

Example of Max Pooling

Consider a **4 × 4 feature map**:

$$\begin{bmatrix} 1 & 3 & 2 & 1 \\ 4 & 6 & 5 & 2 \\ 7 & 2 & 8 & 3 \\ 2 & 4 & 1 & 5 \end{bmatrix}$$

Apply **2 × 2 max pooling with stride = 2**.

Step 1: First Region

$$\begin{bmatrix} 1 & 3 \\ 4 & 6 \end{bmatrix}$$

Maximum value = **6**

Step 2: Second Region

$$\begin{bmatrix} 2 & 1 \\ 5 & 2 \end{bmatrix}$$

Maximum value = **5**

Step 3: Third Region

$$\begin{bmatrix} 7 & 2 \\ 2 & 4 \end{bmatrix}$$

Maximum value = **7**

Step 4: Fourth Region

$$\begin{bmatrix} 8 & 3 \\ 1 & 5 \end{bmatrix}$$

Maximum value = **8**

Output After Max Pooling

$$\begin{bmatrix} 6 & 5 \\ 7 & 8 \end{bmatrix}$$

Thus, the **4×4 feature map becomes 2×2**, reducing dimensionality.

Example of Average Pooling

Using the same 2×2 region:

$$\begin{bmatrix} 1 & 3 \\ 4 & 6 \end{bmatrix}$$

Average value:

$$(1 + 3 + 4 + 6)/4 = 3.5$$

So average pooling outputs **3.5** for that region.

Advantages of Pooling

1. **Reduces overfitting** by decreasing parameters.

2. **Improves computational efficiency**.
3. **Provides translation invariance** to small shifts in input images.
4. **Highlights important features** such as edges and textures.

Applications

Pooling is widely used in CNN models for:

- Image classification
- Object detection
- Face recognition
- Medical image analysis
- Autonomous driving systems

✅ **Conclusion**

Pooling plays an important role in CNNs by reducing feature map dimensions while retaining important information. Max pooling and average pooling are the most commonly used pooling techniques in deep learning models.

3. Explain an expression for Unshared convolution with explanation and explain Tiled convolution.(13marks)

Unshared Convolution and Tiled Convolution

Introduction

In standard convolutional neural networks, the same filter (kernel) is applied across the entire input image. This is called **weight sharing**. However, in some cases different filters are used for different regions of the image. This leads to **Unshared Convolution** and **Tiled Convolution**, which are variations of the convolution operation used to improve feature learning.

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1. Unshared Convolution

Definition

Unshared convolution is a type of convolution where **different filters are applied to different spatial locations of the input** instead of using the same filter everywhere.

Unlike standard convolution, **weights are not shared across the entire input image**.

Expression for Unshared Convolution

In standard convolution:

$$y(i,j) = \sum_m \sum_n x(i+m, j+n)\, w(m,n)$$

In **unshared convolution**, the weights depend on the spatial position.

$$y(i,j) = \sum_m \sum_n x(i+m, j+n)\; w_{ij}(m,n)$$

Where:

- x= input image
- $y(i,j)$= output feature at position (i,j)
- w_{ij}= different kernel for each location
- m, n= kernel indices

This means **each spatial position has its own filter weights**.

Explanation

In unshared convolution, the network learns **different filters for different regions of the image**. For example, the top-left region may use one filter while the bottom-right region uses another.

This approach allows the network to learn **location-specific features** but increases the number of parameters.

Example

Consider an image divided into different regions:

- Region 1 → Filter A
- Region 2 → Filter B
- Region 3 → Filter C

Each region uses a **different convolution filter**, producing different feature maps.

Advantages

1. Captures **location-specific patterns**.
2. Useful when image features depend on **spatial location**.
3. Provides more flexibility than standard convolution.

Disadvantages

1. Large number of parameters.
2. Higher computational cost.
3. Increased risk of overfitting.

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2. Tiled Convolution

Definition

Tiled convolution is a technique where **filters are shared only among small groups of spatial locations instead of the entire image**.

It is a compromise between **fully shared convolution and unshared convolution**.

Explanation

In tiled convolution, the feature map is divided into **tiles**, and filters are shared within each tile.

For example:

- Tile 1 → Filter A
- Tile 2 → Filter B
- Tile 3 → Filter A (repeated pattern)

Thus, weights are **periodically repeated across the input**.

Working of Tiled Convolution

1. The input image is divided into small regions called **tiles**.
2. A set of filters is assigned to each tile.
3. The filters repeat in a **regular pattern across the image**.
4. Convolution is applied separately within each tile.

Example

If the tile size is **2**, filters repeat every two positions:

Filter pattern:

A B
A B
A B

This means filters **A and B alternate across the image**.

Advantages

1. Reduces number of parameters compared to unshared convolution.
2. Maintains some **spatial variability**.
3. Improves learning of **complex patterns**.

Applications

- Image recognition
- Object detection
- Feature learning in deep CNNs
- Unsupervised feature learning

Conclusion

Unshared convolution allows different filters for each spatial location, enabling location-specific feature extraction but increasing the number of parameters. Tiled convolution reduces parameters by sharing filters across repeated tiles while still allowing spatial variation. Both methods extend standard convolution to improve feature representation in deep learning models.

4. Explain in detail the variants of the Basic Convolution Function. (13marks)

Variants of the Basic Convolution Function

Introduction

In Convolutional Neural Networks (CNNs), the **basic convolution operation** extracts features from input data such as images. To improve efficiency, performance, and flexibility, several **variants of the basic convolution function** are used. These variants modify how convolution is applied to the input data.

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1. Strided Convolution

Definition

Strided convolution is a convolution operation where the **filter moves more than one step at a time** across the input.

Explanation

In standard convolution the stride is **1**, meaning the filter moves one pixel at a time.
In strided convolution the stride may be **2 or more**, which reduces the size of the output feature map.

Example

If stride = 2, the filter skips one pixel each time it moves.

Advantages

- Reduces feature map size
- Reduces computation
- Often replaces pooling layers

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2. Zero Padding Convolution

Definition

Zero padding adds **extra rows and columns filled with zeros around the input image** before applying convolution.

Explanation

Without padding, the output feature map becomes smaller than the input.
Padding helps **preserve spatial dimensions** and allows filters to process edge pixels.

Example

A **5×5 image with 3×3 filter** normally gives a **3×3 output**.
With padding, the output can remain **5×5**.

Advantages

- Maintains image size
- Improves feature extraction at edges

- Prevents loss of border information

---

3. Dilated Convolution

Definition

Dilated convolution (also called **atrous convolution**) increases the **spacing between kernel elements**.

Explanation

Instead of applying the filter to adjacent pixels, it skips pixels with a fixed gap.

Example dilation rate = 2:

Normal kernel
1 1 1

Dilated kernel
1 0 1 0 1

Advantages

- Larger receptive field
- Captures wider context
- Useful in segmentation tasks

---

4. Transposed Convolution

Definition

Transposed convolution (also called **deconvolution**) is used to **increase the spatial size of feature maps**.

Explanation

While normal convolution reduces the feature map size, transposed convolution performs **upsampling**.

Applications

- Image generation
- Image segmentation
- Generative models

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5. Separable Convolution

Definition

Separable convolution divides convolution into **multiple smaller operations**.

Two types:

- Depthwise separable convolution
- Spatial separable convolution

Explanation

Instead of applying a full filter to all channels, convolution is separated into **channel-wise operations** followed by pointwise convolution.

Advantages

- Reduces computation
- Reduces parameters
- Used in efficient CNN models like MobileNet

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6. 1×1 Convolution

Definition

A **1×1 convolution** uses a filter of size **1×1**.

Explanation

It processes each pixel across channels and is mainly used to **change the number of channels** in feature maps.

Applications

- Dimensionality reduction
- Feature transformation
- Used in architectures like GoogleNet

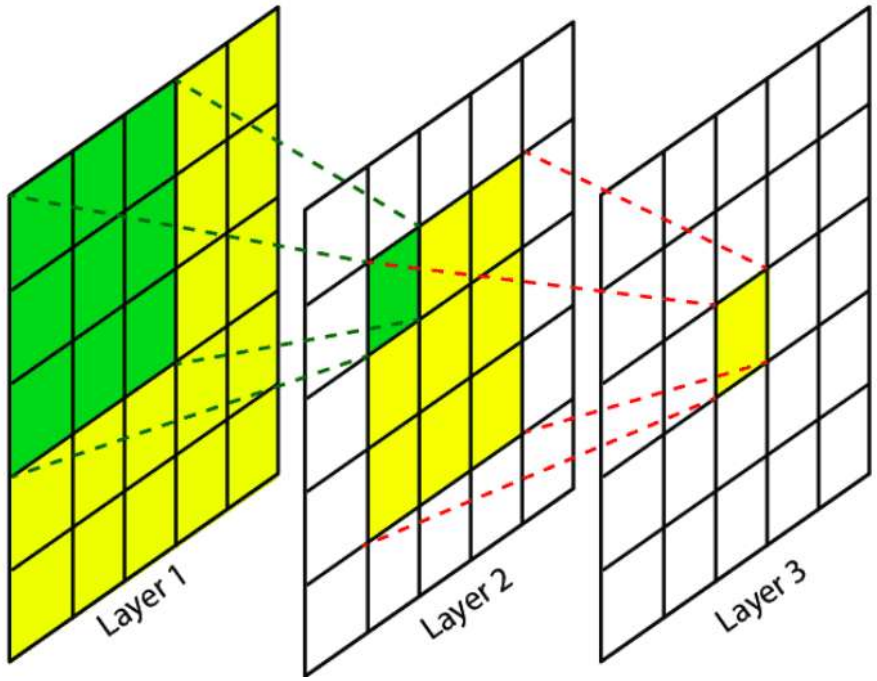
Comparison of Variants

Variant	Purpose
Strided Convolution	Reduces feature map size
Zero Padding	Maintains spatial dimensions
Dilated Convolution	Expands receptive field
Transposed Convolution	Upsamples feature maps
Separable Convolution	Reduces computation
1×1 Convolution	Adjusts channel depth

PART – C
UNIT – 3
1. Construct a graphical demonstration for sparse connectivity and explain it in detail. (15 marks)

Sparse Connectivity in Convolutional Neural Networks (CNN)

Graphical Demonstration



1. Introduction

In **Convolutional Neural Networks (CNNs)**, **sparse connectivity** means that each neuron is connected only to a **small region of the input** rather than to every neuron in the previous layer.

This small region is called the **local receptive field**.

In contrast, in a **fully connected neural network**, every neuron is connected to **all input neurons**, which increases computation and parameters.

Graphical Demonstration (Simple Example)

Input Image (5 × 5)

I1 I2 I3 I4 I5
I6 I7 I8 I9 I10
I11 I12 I13 I14 I15
I16 I17 I18 I19 I20
I21 I22 I23 I24 I25

Convolution Filter (3 × 3 Kernel)

W1 W2 W3
W4 W5 W6
W7 W8 W9

Sparse Connection

The neuron in the feature map connects **only to a small part of the input**:

Input Region Used

I1 I2 I3
I6 I7 I8
I11 I12 I13

Instead of connecting to **all 25 input neurons**, the convolution neuron uses **only 9 inputs**.

Output neuron calculation:

Output = (I1×W1 + I2×W2 + I3×W3
+ I6×W4 + I7×W5 + I8×W6
+ I11×W7 + I12×W8 + I13×W9)

This produces **one value in the feature map**.

Sliding Window (Convolution Operation)

The kernel moves across the image.

Example positions:

Step 1 Step 2
I1 I2 I3 I2 I3 I4
I6 I7 I8 I7 I8 I9
I11 I12 I13 I12 I13 I14

Each step generates **one output neuron**.

Thus the feature map becomes:

O1 O2 O3
O4 O5 O6
O7 O8 O9

Why Sparse Connectivity is Important

1. Reduces Number of Parameters

Example comparison:

Fully Connected Layer

25 inputs × 10 neurons = 250 weights

Convolution Layer

3×3 kernel = 9 weights only

Even if applied across the whole image, **the same 9 weights are reused**.

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2. Captures Local Patterns

Images contain **local features** such as:

- Edges
- Corners
- Textures
- Shapes

Sparse connectivity allows CNNs to focus on **local spatial relationships**.

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3. Reduces Computational Cost

Because each neuron processes **small input patches**, the computation becomes faster and memory usage decreases.

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4. Enables Deep Architectures

Since the number of parameters is smaller, CNNs can have **many layers**, enabling deep learning models like:

- ResNet
- VGG
- AlexNet

Comparison with Fully Connected Network

Feature	Fully Connected	Sparse Connectivity (CNN)
Connections	All neurons connected	Only local region connected
Parameters	Very high	Very low
Efficiency	Slow	Fast
Feature detection	Poor spatial understanding	Excellent for images

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Real-World Example

When detecting a **cat in an image**:

1. First layer detects **edges**
2. Second layer detects **textures**

3. Third layer detects **parts (ears, eyes)**
4. Final layer detects **complete object**

Sparse connectivity helps the network **learn features hierarchically**.

Conclusion

Sparse connectivity is a key concept in CNNs where each neuron connects only to a **small local region of the input instead of the entire input layer**. This reduces parameters, improves computational efficiency, and enables CNNs to effectively learn spatial features in images.

UNIT – 4

1. Examine Unfolding Computational Graphs and Bidirectional RNNs. (13)

Unfolding Computational Graphs and Bidirectional RNNs (13 Marks)

Introduction

Sequence data such as **text, speech, and time-series data** require models that can remember previous information. **Recurrent Neural Networks (RNNs)** are used for this purpose. Two important concepts in RNNs are **Unfolding Computational Graphs** and **Bidirectional RNNs**, which help the network learn from sequential data effectively.

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1. Unfolding Computational Graphs

Definition

Unfolding a computational graph means **expanding a recurrent neural network across different time steps** to show how data flows through the sequence.

Normally an RNN has a **loop structure**, but when unfolded it becomes a **chain of repeated networks**.

Working

For an input sequence:

x1, x2, x3, x4

The network processes each input step by step.

x1 → h1 → y1
x2 → h2 → y2
x3 → h3 → y3
x4 → h4 → y4

Each hidden state depends on:

- Current input
- Previous hidden state

Formula

Hidden state calculation:

$$h_t = f(W_h h_{t-1} + W_x x_t)$$

Where

- x_t= input at time t

- h_t= hidden state
- W_h, W_x= weight matrices
- f= activation function

Advantages

1. Helps understand how RNN works over time
2. Makes it easier to train the network
3. Used in **Backpropagation Through Time (BPTT)**
4. Shows dependency between sequence elements

Applications

- Speech recognition
- Language modeling
- Time-series prediction
- Text generation

2. Bidirectional Recurrent Neural Networks (Bi-RNN)

Definition

A **Bidirectional RNN** processes sequence data in **both forward and backward directions** so that the model can use **past and future information**.

Structure

It has two RNN layers:

1. **Forward RNN** – processes sequence from start to end
2. **Backward RNN** – processes sequence from end to start

Forward: x1 → x2 → x3 → x4
Backward: x1 ← x2 ← x3 ← x4

Both outputs are combined to produce the final result.

Advantages

1. Uses information from **both directions**
2. Improves accuracy in sequence tasks
3. Gives better context understanding

Example

Sentence:

The bank of the river

To understand the word **bank**, the model needs information from both sides.
Bidirectional RNN helps capture this context.

Applications

- Natural Language Processing (NLP)
- Speech recognition
- Machine translation
- Named entity recognition

1. Example for Unfolding Computational Graph

Example: Predicting the next word in a sentence

Sentence:

I love deep learning

The RNN reads one word at a time.

Unfolded computational graph:

Time 1: I → h1 → Output1
Time 2: love → h2 → Output2
Time 3: deep → h3 → Output3
Time 4: learning → h4 → Output4

Explanation:

- The model reads **"I"** and creates hidden state **h1**.
- Then it reads **"love"**, using **previous memory h1**.
- This continues for the entire sequence.
- Each step depends on the **previous hidden state**.

So unfolding shows how the **RNN processes the sequence step by step over time**.

2. Example for Bidirectional RNN

Example: Understanding the meaning of a word in a sentence

Sentence:

The bank of the river is beautiful

The word **bank** can mean:

- financial bank
- river bank

To understand the correct meaning, the model needs **both previous and next words**.

Bidirectional RNN processes:

Forward direction:

The → bank → of → the → river

Backward direction:

river → the → of → bank → The

The model combines both directions to understand that **bank refers to river bank**, not a financial bank.

2. Evaluate Deep Recurrent Networks in detail.(13)

Evaluate Deep Recurrent Networks (13 Marks)

Introduction

Deep Recurrent Networks are an extension of **Recurrent Neural Networks (RNNs)** that contain **multiple hidden layers**.
While a normal RNN has only one hidden layer, a deep RNN stacks several recurrent layers to learn **more complex patterns in sequential data**.

Deep recurrent networks are widely used in **speech recognition, machine translation, and natural language processing**.

Structure of Deep Recurrent Networks

A deep recurrent network contains **multiple RNN layers connected one above another**.

Example structure:

Input Sequence
↓
RNN Layer 1
↓
RNN Layer 2
↓
RNN Layer 3
↓
Output Layer

Each layer learns different levels of features from the sequence.

Working of Deep Recurrent Networks

1. The input sequence is given to the **first RNN layer**.
2. The first layer processes the sequence and produces hidden states.
3. These hidden states become the **input to the next RNN layer**.
4. This process continues through multiple layers.
5. The final layer produces the output.

Thus, information flows **both across time steps and across layers**.

Mathematical Representation

Hidden state calculation:

$$h_t^l = f(W^l h_{t-1}^l + U^l h_t^{l-1})$$

Where:

- h_t^l= hidden state at layer $l$and time t
- h_{t-1}^l= previous time step state
- h_t^{l-1}= output from lower layer
- W^l, U^l= weight matrices
- f= activation function

Advantages of Deep Recurrent Networks

1. **Better feature learning**
   Multiple layers capture complex patterns.
2. **Improved sequence modeling**
   Works well for long and complicated sequences.
3. **Higher accuracy**
   Performs better than shallow RNNs.
4. **Hierarchical representation**
   Different layers learn different levels of information.

Disadvantages

1. **High computational cost**
   More layers require more processing.
2. **Training difficulty**
   Problems like **vanishing gradient** may occur.
3. **Long training time**

Example: Machine Translation using Deep Recurrent Networks

Machine translation converts text from **one language to another language**.

Example

Input sentence (English):

I am a student

Output sentence (French):

Je suis étudiant

Working of Deep Recurrent Network

The sentence is processed **word by word** through multiple recurrent layers.

Input Words
I → am → a → student

↓
RNN Layer 1 (learns basic word features)
↓
RNN Layer 2 (learns sentence structure)
↓
RNN Layer 3 (understands meaning)
↓
Output Sentence
Je → suis → étudiant

Explanation

- **Layer 1** learns basic word patterns.
- **Layer 2** understands grammar and sentence structure.
- **Layer 3** captures the full meaning of the sentence.

Finally, the network generates the translated sentence.

Why Deep RNN Works Well Here

Deep recurrent networks can:

- Understand **long sentences**
- Capture **context between words**
- Learn **complex language patterns**

This makes them suitable for **translation systems like Google Translate**.

Applications of Deep Recurrent Networks

- Speech recognition
- Machine translation
- Text generation
- Language modeling
- Video analysis
- Time-series prediction

Conclusion

Deep Recurrent Networks extend traditional RNNs by using **multiple recurrent layers**. This allows the network to learn **complex sequential patterns and hierarchical features**, making them powerful for applications such as speech recognition, machine translation, and natural language processing.

3. Illustrate Encoder-Decoder sequence-to-sequence Architecture. (13)

Illustrate Encoder–Decoder Sequence-to-Sequence Architecture (13 Marks)

Introduction

The **Encoder–Decoder architecture** is a neural network model used to convert one sequence into another sequence. It is also called **Sequence-to-Sequence (Seq2Seq) architecture**.

This model is widely used in tasks where the **input and output lengths may be different**, such as machine translation, text summarization, and speech recognition.

Encoder–Decoder Architecture

The architecture consists of two main parts:

1. **Encoder**
2. **Decoder**

Input Sequence → Encoder → Context Vector → Decoder → Output Sequence

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1. Encoder

The **encoder** reads the input sequence step by step and converts it into a fixed-length vector called the **context vector**.

Example input sentence:

I love AI

Processing:

I → h1
love → h2
AI → h3

The final hidden state **h3** becomes the **context vector**, which represents the meaning of the entire sentence.

---

2. Context Vector

The **context vector** is the compressed representation of the input sequence.

It stores important information about the input and passes it to the decoder.

Context Vector = final hidden state of encoder

---

3. Decoder

The **decoder** takes the context vector and generates the output sequence step by step.

Example output sentence (French):

Je aime IA

Decoder process:

Context → y1 → y2 → y3

Each output word is generated using the previous output and hidden state.

Working of Encoder-Decoder Model

1. The **encoder reads the entire input sequence**.
2. It converts the sequence into a **context vector**.
3. The **decoder receives the context vector**.
4. The decoder generates the **output sequence one element at a time**.

Example: Machine Translation

Input (English):

I am happy

Encoder converts it into a context vector.

Decoder generates output (French):

Je suis heureux

Thus, the encoder understands the meaning of the input sentence and the decoder produces the translated sentence.

Advantages

1. Works well for **sequence-to-sequence tasks**
2. Can handle **variable length inputs and outputs**
3. Learns **context and relationships between words**

Applications

- Machine translation
- Text summarization
- Chatbots
- Speech recognition
- Question answering systems

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4. Explain Leaky Units and Other Strategies for Multiple Time Scales. (13)

Explain Leaky Units and Other Strategies for Multiple Time Scales (13 Marks)

Introduction

In sequence modeling, neural networks must capture **dependencies over different time scales**. Some information is needed only for short periods, while other information must be remembered for a long time.
To solve this problem, techniques such as **Leaky Units and other strategies for multiple time scales** are used in Recurrent Neural Networks (RNNs).

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1. Leaky Units

Definition

Leaky units are special neurons used in RNNs that allow the hidden state to **retain part of the previous information** while updating with new input.

Instead of completely replacing the previous hidden state, leaky units **slowly update the hidden state**, allowing the network to remember information for longer periods.

Working

In a normal RNN, the hidden state is fully replaced at each time step.
In leaky units, the hidden state is updated gradually.

Formula:

$$h_t = (1 - \alpha)h_{t-1} + \alpha f(Wx_t + Uh_{t-1})$$

Where:

- h_t= current hidden state
- h_{t-1}= previous hidden state
- x_t= current input
- α= leak rate (controls how fast the state changes)

If **α is small**, the network remembers information for a longer time.

Advantages

1. Helps remember **long-term information**
2. Reduces **vanishing gradient problem**
3. Allows smoother updates of hidden states

Example

Consider **temperature prediction over time**.

Day 1 → 30°C
Day 2 → 31°C
Day 3 → 32°C

Temperature changes slowly, so the network should remember previous values.
Leaky units allow the model to **retain previous information gradually**.

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2. Strategies for Multiple Time Scales

Different parts of sequence data may change at **different speeds**.
Some features change quickly, while others change slowly.

Several strategies are used to handle multiple time scales.

Strategy 1: Using Multiple Hidden Layers

Different layers capture different time scales.

Input Sequence
↓
Layer 1 → captures short-term patterns
↓
Layer 2 → captures medium-term patterns
↓
Layer 3 → captures long-term patterns

This helps the network understand **both short and long dependencies**.

Strategy 2: Clockwork RNN

In this approach, hidden units are divided into **modules that operate at different time intervals**.

Example:

Module 1 → updates every time step
Module 2 → updates every 2 steps
Module 3 → updates every 4 steps

Thus, slower modules capture **long-term information**, while faster modules capture **short-term patterns**.

Strategy 3: Hierarchical RNN

Hierarchical RNNs process sequences at **different levels**.

Example in language:

Character level → Word level → Sentence level

This allows the model to learn information at **multiple time scales**.

Applications

Leaky units and multi-time-scale strategies are used in:

- Speech recognition
- Language modeling
- Video analysis
- Time-series forecasting
- Natural language processing

Conclusion

Leaky units help RNNs remember information by slowly updating hidden states. Other strategies such as **multiple hidden layers, clockwork RNN, and hierarchical RNN** allow networks to capture patterns at different time scales. These techniques improve the ability of neural networks to handle **long-term dependencies in sequential data**.

1. Example for Leaky Units

Example: Weather Prediction

Suppose we want to predict **temperature over several days**.

Day 1 → 30°C
Day 2 → 31°C
Day 3 → 32°C
Day 4 → 33°C

Temperature usually changes **slowly over time**.
So the neural network should **remember previous temperature values**.

With **Leaky Units**:

- The hidden state is **not completely replaced**
- It keeps **some information from the previous day**

Example:

Previous temperature memory + New temperature input → Updated prediction

Thus, leaky units allow the network to **retain past information gradually**.

---

2. Example for Multiple Time Scales

Example: Language Processing

Consider the sentence:

I am studying deep learning

Different information occurs at **different time scales**.

Time Scale	Example
Character level	I, a, m
Word level	studying
Sentence level	full sentence meaning

A **multi-time-scale RNN** can learn:

Layer 1 → character patterns
Layer 2 → word patterns
Layer 3 → sentence meaning

Thus, the network captures **short-term and long-term dependencies**.

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3. Example: Video Activity Recognition

In video analysis:

Frame 1 → walking
Frame 2 → walking
Frame 3 → running

Some actions change **quickly**, while others change **slowly**.

A multi-time-scale model can learn:

Fast layer → frame changes
Slow layer → overall activity

✅ **Short line you can add in exam**

Leaky units and multi-time-scale strategies help neural networks **remember long-term information and capture patterns occurring at different speeds in sequential data**.

5. Explain Optimization for Long-Term Dependencies. (13)with example

Explain Optimization for Long-Term Dependencies (13 Marks)

Introduction

In sequence modeling using **Recurrent Neural Networks (RNNs)**, the network must remember information from **earlier time steps**. Sometimes important information appears far earlier in the sequence. Learning such relationships is called **long-term dependencies**.

However, training RNNs for long sequences is difficult due to problems like **vanishing gradients and exploding gradients**. Therefore, several **optimization techniques** are used to handle long-term dependencies effectively.

Problems in Long-Term Dependencies

1. Vanishing Gradient Problem

During training, gradients become **very small** when propagated through many time steps. As a result, the network **cannot learn information from earlier inputs**.

2. Exploding Gradient Problem

Gradients become **very large**, causing unstable learning and large weight updates.

These problems make it difficult for RNNs to learn long-term relationships.

Optimization Techniques for Long-Term Dependencies

1. Long Short-Term Memory (LSTM)

LSTM is a special type of RNN designed to **store information for long periods** using memory cells.

It uses **three gates**:

- **Forget Gate** – removes unnecessary information
- **Input Gate** – adds new information
- **Output Gate** – decides what information to output

This gating mechanism helps maintain **long-term memory**.

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2. Gated Recurrent Unit (GRU)

GRU is a simplified version of LSTM.

It has two gates:

- **Update Gate**
- **Reset Gate**

GRUs help the network **remember important information and forget irrelevant data**, improving learning of long sequences.

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3. Gradient Clipping

Gradient clipping is used to prevent **exploding gradients**.

If gradients become too large, they are **limited to a maximum threshold**.

Example:

If gradient > threshold
then gradient = threshold

This stabilizes training.

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4. Better Weight Initialization

Proper initialization of weights helps prevent gradients from becoming too large or too small.

It improves the **training stability of RNNs**.

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5. Skip Connections

Skip connections allow information to **skip certain time steps**, helping gradients flow more easily during training.

This improves learning of long-term relationships.

Example

Example: Language Processing

Consider the sentence:

I grew up in France, and I speak fluent French.

To understand why the person speaks French, the model must remember the word **France**, which appeared earlier in the sentence.

A simple RNN may forget this information due to the **vanishing gradient problem**.

Using **LSTM or GRU**, the model can remember important information for a long time and correctly understand the sentence.

Applications

Optimization for long-term dependencies is used in:

- Speech recognition
- Machine translation
- Text generation
- Language modeling

- Time-series prediction

Conclusion

Long-term dependencies are difficult for traditional RNNs because of vanishing and exploding gradients. Optimization techniques such as **LSTM, GRU, gradient clipping, better weight initialization, and skip connections** help neural networks learn long-term relationships effectively in sequential data.